ILIA RAKHLEVSKI

Senior C/C++ Software Engineer

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Professional Summary

Senior Software Engineer with 20 years of overall software development experience, specializing in C/C++ development, long-lived codebase maintenance, and system modernization. Skilled in analysing and extending complex legacy systems across domains, from embedded devices to desktop applications, including industrial and mission-critical environments. Experienced in stabilising fragile software, improving performance, and reducing technical debt in large-scale systems. Brings a pragmatic, engineering-oriented mindset to modernising existing systems while preserving functionality. Solid background in cross-platform development (Windows/Linux), debugging, and system-level design.

Key Skills

Primary Expertise

- **Languages**: C, C++
- **EDA/CAD Systems**: development and maintenance of complex PCB/DFM tools (Valor NPI), large-scale legacy codebases
- **Legacy Systems**: long-term maintenance, refactoring, performance tuning, debugging large, undocumented code
- **Development Environments**: Visual Studio (C/C++), cross-platform development (Windows/Linux)
- **Version Control**: Git, SVN
- **Debugging & Profiling Tools**: Visual Studio Debugger, custom logging, performance profilers

Relevant Industry Experience

- **Embedded Development**: bare-metal STM32 (ARM Cortex-M3/M4), MSP430, FreeRTOS, device drivers, UART/SPI/I2C/ADC, signal analysis
- **GUI Development**: WinForms, WPF, Qt/QML (older experience), remote control interfaces for robotics
- **Security & Robotics**: sensor firmware, robotic control, video streaming
- **Languages & Tools**: Python 3 (data processing, developing automated testing suites for data verification and results analysis), C#, .NET
- **Multithreading**: basic experience with concurrent and parallel programming

Education & Project Work

- **Machine Learning & Data Science**: Python, R, Keras, PyTorch, ResNet+LSTM, Faster R-CNN (academic projects)
- **Linux & Embedded Linux**: mini2440 EVB (evaluation board), user/kernel-space development (Real Time College)
- **Other**: RStudio, Spyder and IPython, IAR Embedded Workbench, STM32CubeMX

Professional Experience:

Siemens EDA (formerly Mentor Graphics) — Senior Software Engineer

Israel, March 2020 – February 2026

Siemens EDA develops advanced software tools for electronic design automation.
Main product: Valor NPI — a complex system for DFM (Design for Manufacturing) analysis of printed circuit boards (PCBs).

- Acted as domain expert in two critical areas:
  — Import and conversion of external PCB designs (e.g. Cadence Allegro, Gerber, Excellon, DXF) into the internal ODB++ format
  — Netlist analysis, including logical validation and complex cross-layer consistency checks to ensure data integrity in imported PCB designs.
- Modernised and maintained large, long-lived C/C++ codebases
- Resolved performance issues and improved overall system stability
- Investigated and debugged complex automated regression test failures across multiple functional areas
- Tools & technologies: C, C++, Python, Visual Studio, Git/SVN, Windows, Linux

Real Time Group (Outsourced Projects) — Embedded Software Engineer

Israel, November 2017 – February 2020

Worked on embedded firmware projects for external clients in the security and industrial sectors, focusing on bare-metal development, low-level drivers, and hardware integration.

Client: Magal Security Systems
Advanced fence-mounted perimeter intrusion detection system

- Developed firmware for sensors detecting vibrations, movement, and physical tampering
- Applied signal processing and machine learning techniques for event classification
- Designed and implemented new sensor firmware architecture from scratch, including signal acquisition and control logic
- Performed optimisation and debugging on performance-critical code in resource-limited environments
- Tools & technologies: ARM Cortex-M4 (NXP Kinetis, STM32), MSP430, UART, SPI, I2C, ADC, DMA, IAR Embedded Workbench, FreeRTOS, STM32CubeMX HAL, C, C#, WPF, GUI, Python, R, Spyder and IPython, multithreading, Git, Windows 10

Client: Electra A/C
Control unit for central air conditioning systems

- Acted as Technical Lead, providing code reviews, technical guidance, and hands-on software development within the project team
- Developed low-level drivers for outdoor unit control: stepper motor, compressor, thermistor sensing, communication
- Diagnosed and fixed critical issues in legacy embedded codebase, significantly improving system reliability and maintainability
- Built diagnostic PC tool for real-time system monitoring and testing
- Tools & technologies: ARM Cortex-M3 (STM32), Saleae Logic Analyzer, UART, SPI, ADC, embedded HMI, IAR Embedded Workbench, C, STM32 Standard Peripheral Library, Visual Studio 2017, C++, MFC, Windows 7, SVN

Roboteam Ltd. — Software Engineer

Israel, January 2016 – November 2017

Roboteam develops unmanned robotic systems used in security, rescue, and field operations.
Worked on control software for robotic platforms, focusing on real-time communication, telemetry, and video integration.

- Designed and developed desktop control software, including from-scratch projects as well as extensions to existing systems (UI, communication, coordination logic)
- Built user interfaces for remote control, telemetry, and real-time video streaming (using Live555 media streaming library)
- Implemented communication modules using OpenJAUS protocol and custom interfaces, with multithreaded processing for real-time data handling
- Collaborated with hardware and QA teams to ensure reliable performance in field conditions
- Tools & technologies: C++, Qt/QML, Win32 API, MFC, C#, WPF/XAML, Visual Studio, multithreading, design patterns, Windows 7/10, Linux (Ubuntu), Git

PTC Inc. — Software Development Engineer

Israel, June 2011 – December 2015

PTC is a U.S.-based software company focused on CAD and product lifecycle management tools.
Worked on the Creo suite of 2D CAD products, including Creo Schematics, Creo Sketch, and Creo Layout.

- Maintained and enhanced large-scale legacy C/C++ codebases across multiple products
- Implemented GUI features and contributed to code modernisation and modularisation
- Optimised performance and resolved memory leaks and corruption issues in production builds
- Investigated and resolved failures in automated regression tests, including routine and complex issues
- Worked on both Windows and Unix (Sun Solaris) environments

Tools & technologies: C, C++, Java, MEDUSA CAD platform, Visual Studio, Unix (Sun Solaris), Jenkins, Cygwin, Performance Profiler, Application Verifier, internal frameworks and tools of PTC, Windows 7

C-True Ltd. — Software and QA Engineer

Israel, October 2008 – June 2011

C-True developed face recognition systems for access control and identity verification.
Worked across the full development stack, including client-server communication, GUI, testing, and data management.

- Developed software modules for face recognition systems, including client and server-side components
- Built internal tools for automated and manual testing of biometric algorithms and system behaviour
- Designed a tracking system for products, orders, and bugs (internal bug tracker with reporting features)
- Performed debugging, UI development, and integration of data exchange between system components
- Tools & technologies: C++, Win32 API, MFC, C#, .NET 3.5/4, WinForms, WPF, WCF, ADO.NET, XML-RPC, SQL, Visual Studio, design patterns, multithreading, Windows Server 2008

MBE-Simulation Ltd. — Software Programmer

Israel, February 2007 – September 2008

Worked on simulation software for modelling management and operational processes.
Developed desktop applications with interactive user interfaces and visualisation logic.

- Designed and implemented simulation components in C++ with focus on user interaction and visual feedback
- Built GUI elements for configuring and monitoring simulated processes
- Tools & technologies: C++, MFC, Win32 API, GDI, Visual Studio, Windows XP/Vista, HTML (for embedded UI help)

Segnetron Israel Ltd. — Software Programmer

Israel, December 2000 – November 2001

Worked on embedded and desktop software for medical and measurement devices.

- Developed desktop software for camera-based medical image processing
- Developed firmware for laser-based distance measurement and oxygen level monitoring using embedded microcontrollers (8031/51, COP8)
- Tools & technologies: C++, MFC, Win32 API, GDI, Assembly, Embedded C, Visual Studio, Windows 98

Other Professional Experience

Israel, 2002 – 2006

Worked in various non-technical roles during this period.

Education

M.Sc. in Intelligent Systems
Afeka Tel Aviv Academic College of Engineering, 2018 – 2022 (GPA: 80.95)
Multi-disciplinary M.Sc. program focused on Intelligent Systems, combining Technology, Science, and Engineering aspects of Design, Analysis, and Management of Complex AI-driven Systems.

Key subjects included Statistics, Machine Learning, Deep Learning, Sensors and Signals, Computer Vision, NLP, Speech Processing, Time Series Analysis, Multi-Agent Systems, Autonomous Vehicle and Intelligent Transportation Systems.

Final project: Detecting violence against children in video using deep learning (Python, OpenCV, Keras, ResNet + LSTM)

B.Sc. in Software Engineering
Afeka Tel Aviv Academic College of Engineering, 2005 – 2010 (GPA: 84.36)
Specialisation in software development and information systems
Final project: Von Neumann architecture simulator (C#, .NET 3.5, WinForms)

Practical Software Engineer (Diploma)
College of Management, Haifa, 1998 – 2000 (GPA: 82)
Final project: Scientific calculator for Windows 98 (C++, VC++ 6, MFC)

Courses & Certifications

Technion – Certified Data Science
2023 – 2024
Advanced training in data science and deep learning.
Final projects:

- Machine learning model for matching participants using speed dating dataset (Python, EDA, classification)
- License plate detection using deep learning (PyTorch, Faster R-CNN)

Real Time College — Embedded Linux & Real-Time Development
2016 – 2018

Multi-disciplinary program combining development, integration, and training in embedded systems and real-time software solutions.

Subjects covered:
Embedded C for Bare Metal and Embedded Linux, Linux Administration, RT & Embedded Concepts, ARM Architecture Programming, Networking, Python, Linux Kernel modules and device drivers, Embedded Linux application development, socket communication.

Key Projects:

- Developed servo/stepper motor controller — Bare-metal programming on mini2440 EVB (C)
- Designed and implemented smart home infrastructure — Kernel and user-space development using Ubuntu (PC) and mini2440 EVB (Embedded Linux); C, client/server architecture, multithreading

Sela College – C++ and System Programming Workshops
Completed between 2011 and 2015 (during employment at PTC)

- C++ Debugging and Problem Analysis (16 hours)
- Design Patterns in C++ (32 hours)
- Win32/64 Multithreading and Multiprocessing (24 hours)